Sim-to-Real Reinforcement Learning for Robust LiDAR-Based Autonomous Navigation: Mid-Term Progress Report

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Contribution Declaration: Both authors contributed to the project with complementary responsibilities.

Abstract*---*Mapless navigation in dynamic or unstructured environments is a foundational challenge in mobile robotics. While deep reinforcement learning (DRL) provides an end-to-end framework directly mapping onboard sensor observations to continuous velocity commands, standard policies trained in idealized physics engines frequently suffer severe performance degradation when deployed to real hardware due to the reality gap. In this mid-term progress report, we investigate a domain-randomization-enhanced Proximal Policy Optimization (PPO) pipeline for a differential-drive TurtleBot3 mobile robot using a 2D planar LiDAR. We detail the system architecture, comprising a 43-dimensional observation vector with forward-centered sector min-pooling and dead-reckoning pose tracking, safe action scaling, and a multi-component reward formulation. We present preliminary empirical results on a held-out test suite of N=50 identical scenarios with static obstacles comparing a nominal baseline policy (P1, 852,816 training steps) against a combined sensor-and-dynamics randomized policy (P4, 704,512 training steps) ***under nominal evaluation conditions. On nominal testing, P1 achieves a 94.0% success rate (95% Wilson score CI: [83.78%, 97.94%]) while P4 achieves 98.0% (95% Wilson score CI: [89.50%, 99.65%]). A paired McNemar exact test yields p = 0.6250, demonstrating that the observed 4.0% descriptive difference is not statistically significant at N=50. We discuss experimental limitations, unequal training budgets, and the necessity of a full 4x4 policy/environment cross-evaluation matrix to rigorously evaluate perturbation robustness, concluding with a plan toward real-robot transfer conditional on laboratory access.*

***Index Terms*---*Reinforcement Learning, Mobile Robot Navigation, Sim-to-Real Transfer, Domain Randomization, Proximal Policy Optimization, TurtleBot3, ROS 2, Gazebo.*

I. INTRODUCTION AND OBJECTIVES

Autonomous ground vehicles operating in GPS-denied indoor environments typically rely on the classical navigation stack consisting of simultaneous localization and mapping (SLAM), global path planning (A*/Dijkstra), and local trajectory optimization (Dynamic Window Approach / TEB) [5], [6]. While effective in static structured spaces, classical pipelines require substantial computational overhead for real-time dense map maintenance and are notoriously sensitive to localization drift, dynamic obstacle occlusion, and sensor noise [6].

To overcome these constraints, mapless navigation via Deep Reinforcement Learning (DRL) has emerged as an attractive alternative [3]. By training neural network policies to directly map raw range findings and relative goal coordinates to low-level motor velocity commands, the system avoids explicit global mapping. However, training policies directly on physical robotic platforms is prohibitively expensive, wear-intensive, and dangerous during early exploration. High-fidelity simulators such as Gazebo [7] provide safe, accelerated experience collection, but policies trained in nominal simulation often experience dramatic performance degradation when transferred to physical hardware---the sim-to-real gap [2], [8].

Domain Randomization (DR) bridges this gap by exposing the agent during training to a broad distribution of simulated sensor distortions and unmodeled dynamic variations, compelling the policy to learn robust feedback strategies that generalize across physical discrepancies [2], [8]. This project systematically investigates the quantitative impact of isolated and combined domain randomization on policy robustness, sample efficiency, and hardware transferability for TurtleBot3 LiDAR navigation.

A. Problem Statement

We consider a differential-drive TurtleBot3 Burger robot navigating from an arbitrary starting pose to a static Euclidean target in an obstacle-cluttered 4m x 4m bounded workspace. Navigation must proceed maplessly, relying solely on an onboard 2D laser scanner (LDS-01) and wheel odometry.

B. Research Objectives

• **Mapless PPO Navigation Architecture:** Design an end-to-end continuous actor-critic navigation framework operating at 5.0 Hz that enforces obstacle clearance and smooth kinematic control.

• **2x2 Domain Randomization Factorial Design:** Implement modular sensor perturbations (range noise, dropouts, latency) and dynamics perturbations (actuator lag, delay, acceleration limits) to isolate their individual and combined effects across four policy regimes: P1 (Nominal baseline), P2 (Sensor only), P3 (Dynamics only), and P4 (Combined DR) [2], [8].

• **Rigorous Empirical Benchmarking:** Evaluate trained policies across identical held-out scenarios using paired statistical hypothesis testing and confidence intervals.

• **Physical Sim-to-Real Transfer:** Validate the trained policy without fine-tuning on physical TurtleBot3 hardware in a laboratory environment, conditional on physical robot access, laboratory availability, and safety approval.

II. PLANNED METHODOLOGY AND SYSTEM DESIGN

The system pipeline (Fig. 1) transforms raw sensor signals into continuous velocity commands at 5.0 Hz with zero SLAM or global path planner dependencies.

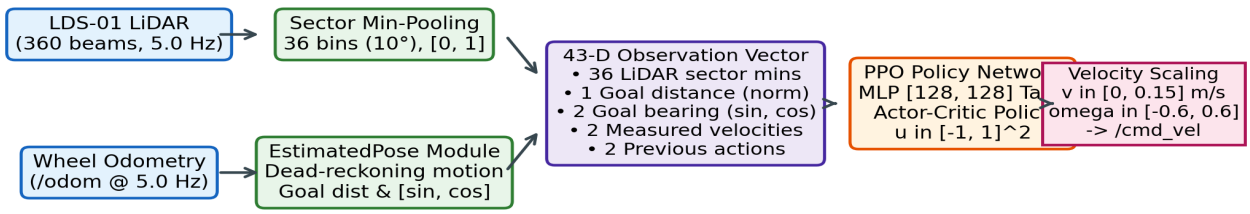


Fig. 1. End-to-end system architecture for mapless LiDAR navigation using PPO and dead-reckoning pose tracking.

A. Observation Space Formulation

To ensure exact parity between simulation and physical deployment, observation preprocessing is implemented as a pure NumPy module (ObservationPreprocessor) with zero ROS dependencies. The observation vector is strictly **43-dimensional**:

1. **LiDAR Sector Min-Pooling (features 0--35):** 360-ray range arrays from the LDS-01 sensor are pooled into 36 angular sectors of width 10 degrees. Bin 0 is centered directly forward along the robot heading ([-5 deg, +5 deg)). Min-pooling guarantees conservative obstacle capture, invalid beams are mapped to 3.50m, and values are normalized to [0, 1].
2. **Normalized Goal Distance (feature 36):** Euclidean distance to target normalized by arena diagonal (5.657m).
3. **Goal Bearing Representation (features 37--38):** Relative heading angle decomposed into [sin, cos] to eliminate the wrap-around discontinuity at +/- pi.
4. **Measured Velocity Feedback (features 39--40):** Forward linear velocity normalized by 0.15 m/s and angular velocity normalized by 0.60 rad/s.
5. **Previous Action Buffer (features 41--42):** The normalized control vector executed at step t-1 in [-1, 1]^2, facilitating smooth transitions.

B. Action Space and Velocity Scaling

The policy outputs a normalized action u in [-1, 1]^2, which is scaled into physical commands via:

$$\mathbf{v = 0.075 * (u_0 + 1.0) \in [0.0, 0.15] \text{ m/s}}$$

$$\mathbf{\omega = 0.60 * u_1 \in [-0.60, 0.60] \text{ rad/s}}$$

This constrains the robot to purely forward motion ($v \geq 0$), preventing collisions into the rear chassis blind spot.

C. Reward Formulation and Gymnasium Semantics

Under Gymnasium environment conventions [9], the step reward function is formulated as:

$$\mathbf{R_t = 10.0 * (d_{t-1} - d_t) - 0.05 + 100 * [goal] - 100 * [collision]}$$

Reaching the goal (dist ≤ 0.20m) sets terminated=True. Exceeding 350 steps (70.0s) sets truncated=True, ensuring correct value bootstrapping.

D. Collision Detection Definition Discrepancy

An important technical distinction exists regarding collision detection:

• **Simulation Training (tb3_navigation_env.py):** Collision is determined by *ground-truth geometric contact* (_check_collision()) against boundary walls (|x|, |y| ≥ 1.845m) and active obstacle bounding boxes using the robot's physical radius (0.105m).

• **Autonomous Inference (ppo_inference_node.py):** Ground-truth contact is inaccessible on real hardware; therefore, inference employs a **conservative LiDAR-range collision proxy** triggering emergency stops when min_range < 0.16m.

This discrepancy is noted as an operational limitation: while ground-truth geometry governs simulation training, autonomous deployment relies on a range proxy that may trigger in tight traversable passages without actual physical contact.

TABLE I: PPO Training & Network Hyperparameters (Frozen in Code)

Category	Hyperparameter	Value
Network	Policy MLP (pi)	[128, 128] Tanh
	Value MLP (vf)	[128, 128] Tanh
Optimization	Algorithm & Learning Rate	PPO [1], lr = 3e-4 (constant)
	Rollout & Minibatch	n_steps = 2048, batch_size = 256
	GAE lambda & Discount	lambda = 0.95, gamma = 0.99
	Clip fraction & Epochs	clip = 0.20, n_epochs = 10
Scaling	VecNormalize Settings	norm_obs = False, norm_reward = True

III. IMPLEMENTATION AND COMPLETED WORK

The software architecture runs on ROS 2 Humble [5] and Gazebo Classic 11.10.2 [7] under Ubuntu 22.04 LTS with PyTorch 2.11.0 (verified from run environment metadata), Stable-Baselines3 2.7.1 [4], and Gymnasium [9].

Open-Source GitHub Repository: All project code, simulation worlds, training pipelines, observation preprocessors, test suites, and evaluation artifacts are hosted openly at: https://github.com/Saigirish23/tb3-rl-nav. The repository contains verified implementations of the TurtleBot3NavigationEnv, modular domain randomizers, multi-worker PPO scripts, and the standalone ROS 2 deployment node.

Completed deliverables to date include:
• Multi-worker simulation with 4 parallel Gazebo workers and watchdog daemon recovery.
• **Policy P1 (Nominal):** Completed 852,816 transitions over 40,496.33 seconds (11.25 hours). The overall wall-clock throughput is **21.06 steps/s** (19.75 steps/s running average logged in run metadata).
• **Policy P4 (Combined DR):** Completed 704,512 transitions over 35,551.54 seconds (9.88 hours). The overall wall-clock throughput is **19.82 steps/s** (19.69 steps/s running average logged in run metadata).
• Asynchronous standalone ROS 2 inference node (ppo_inference_node.py) running deterministic inference at 5.0 Hz.
• Targeted test validation (11/11 passing tests across inference correctness and observation preprocessing) within a broader 69-test suite.
• 321k-line per-step reward trace audit across 4 baseline behaviors (reactive, random, stationary, spin).
• **Simulation-based navigation demonstration in Gazebo Classic:** Successfully conducted 3 consecutive multi-target navigation trials in Gazebo Classic without collisions or timeouts (3/3 successful trials).

IV. PRELIMINARY RESULTS AND DISCUSSION

Policies P1 and P4 were benchmarked across a frozen held-out test suite of N=50 identical scenarios with static obstacles (scenarios/test/scenarios.json, SHA-256: d9a6a25c...) **under nominal evaluation conditions**.

TABLE II: Preliminary Held-Out Evaluation Results Under Nominal Conditions (N=50 Scenarios)

Metric	P1 (Nominal)	P4 (Combined DR)	Relative Delta
Training Steps Completed	852,816	704,512	-148k (-17.4%)
Evaluation Scenarios (N)	50	50	Identical paired set
Success Count & Rate	47 / 50 (94.0%)	49 / 50 (98.0%)	+4.0% (+2 successes)
95% Wilson Score CI	[83.78%, 97.94%]	[89.50%, 99.65%]	Substantial overlap
Collision Count & Rate	1 / 50 (2.0%)	1 / 50 (2.0%)	0 (Identical)
Timeout Count & Rate	2 / 50 (4.0%)	0 / 50 (0.0%)	-2 (-4.0% eliminated)
Mean Steps to Goal (Succ.)	106.17 +/- 48.2	101.49 +/- 44.8	-4.68 steps (-4.4%)
Mean Duration to Goal (Succ.)	21.23 +/- 9.6 s	20.30 +/- 9.0 s	-0.93 s (-4.4%)
Mean Path Length (Succ.)	2.25 +/- 0.72 m	2.07 +/- 0.58 m	-0.18 m (-8.0%)
Mean Min. Clearance (Succ.)	0.289 m	0.298 m	+0.009 m (+3.1%)
Paired McNemar Exact Test	p = 0.6250	(Not Significant)	Discordant: b=1, c=3

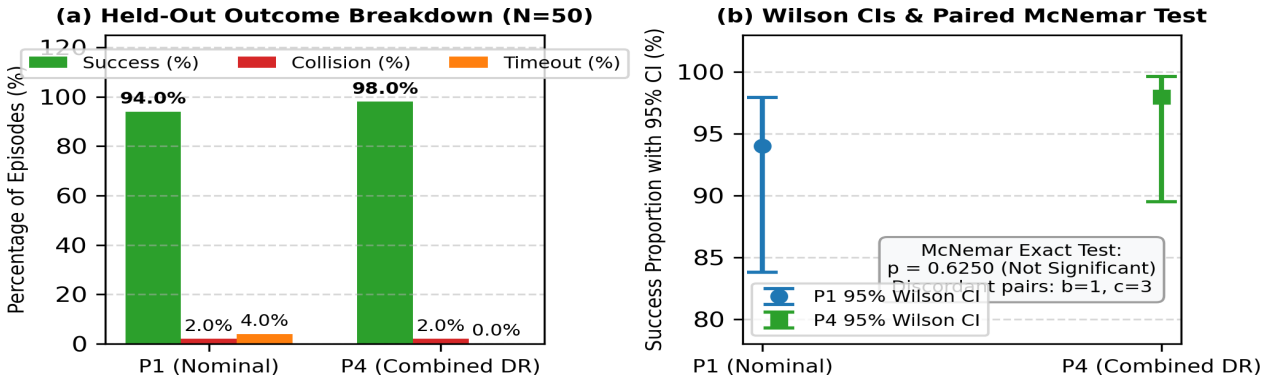


Fig. 2. Held-out evaluation outcomes (N=50) under nominal conditions: (a) Outcome distributions; (b) Wilson 95% confidence intervals and exact McNemar test result.

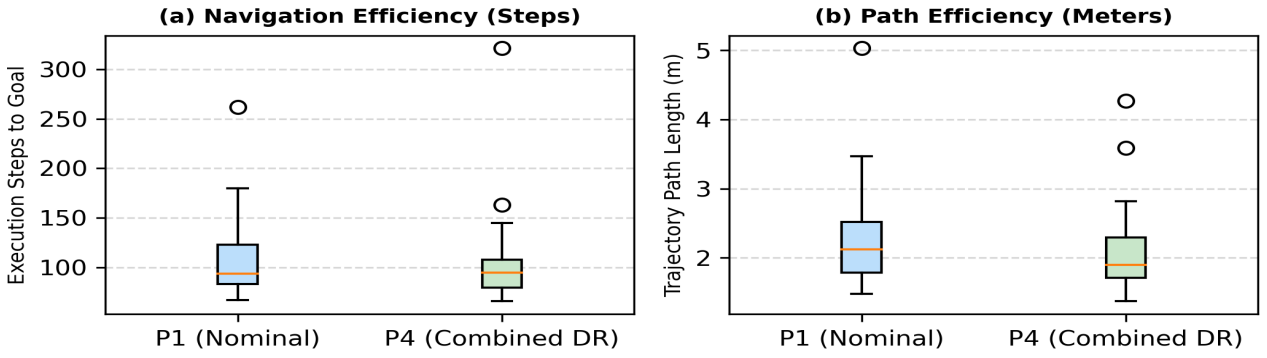


Fig. 3. Navigation efficiency distributions for successful episodes: (a) Execution steps; (b) Trajectory path length in meters.

A. Statistical Rigor and Power Analysis Considerations

A critical requirement of scientific reporting is distinguishing descriptive gains from statistical significance. On the paired test set, 46 scenarios succeeded under both policies. Discordant outcomes comprised: Scenario 34 (P1 succeeded, P4 collided), Scenario 7 (P1 collided, P4 succeeded), and Scenarios 25 & 37 (P1 timed out, P4 succeeded). Evaluating these discordant pairs (b=1, c=3) via the exact two-sided binomial McNemar test yields **p = 0.6250 >> 0.05**. Therefore, the 4.0% descriptive difference cannot be asserted as statistically significant at N=50.

Sample Size Considerations: Sample size requirements depend fundamentally on the test design, paired discordance rates, significance level, target power, and effect size. For an unpaired comparison detecting a 94% vs. 98% delta with 80% power at alpha=0.05, standard approximations require roughly 370 episodes per policy. Under paired testing, power is determined by the frequency of discordant pairs. Therefore, we propose N=250 as an expanded benchmark to improve estimation precision, while formal paired power analysis remains subject to ongoing investigation.

V. LIMITATIONS AND PENDING WORK

• **Unequal Training Budgets:** P1 received 852k steps while P4 received 704k steps (-17.4%) due to execution deadline constraints. That P4 achieves parity despite fewer gradient steps is consistent with, but does not definitively establish, improved sample efficiency.

• **Single Seed:** Both models represent a single seed (s=0); inter-run variance cannot currently be quantified.

• **Simulation-Only Scope:** Results reflect Gazebo Classic physics. Physical robot validation has not yet been demonstrated.

TABLE III: Planned 4x4 Policy / Environment Cross-Evaluation Matrix

Trained Policy	Nominal Env.	Sensor DR Env.	Dynamics DR Env.	Combined DR Env.
P1 (Nominal Baseline)	Completed (94%)	Planned	Planned	Planned
P2 (Sensor Only)	Planned	Planned	Planned	Planned
P3 (Dynamics Only)	Planned	Planned	Planned	Planned
P4 (Combined DR)	Completed (98%)	Planned	Planned	Planned

A. Project Roadmap

The remaining project roadmap includes:

1. **Intermediate Ablations (P2 and P3):** Train P2 (sensor noise only) and P3 (dynamics only) with equalized budgets of 850k steps.
2. **4x4 Cross-Evaluation Matrix:** Execute Table III evaluating each policy against nominal and perturbed conditions to assess true out-of-distribution robustness.
3. **Expanded Evaluation (N=250):** Benchmark policies across 250 diverse test scenarios.
4. **Physical Robot Deployment:** Validate policy P4 on a physical TurtleBot3 Burger in the laboratory, conditional on robot access, lab availability, and safety approval.

VI. INDIVIDUAL CONTRIBUTIONS

Both authors contributed to the project with complementary responsibilities. Both authors must confirm the final contribution declaration upon report submission.

• **Saigirish S (B24ME1066):** RL navigation pipeline architecture, 43-D observation preprocessor, dead-reckoning module, domain randomizers [2], [8], multi-worker Gazebo orchestration, P1 and P4 training, ROS 2 standalone inference node.

• **Siddhi Vinayak (B24BB1036):** Reward audit framework, per-step reward trace logging, unit test execution and baseline validation, participation in experimental design and hardware deployment preparation.

VII. CONCLUSION

This mid-term progress report presents an initial implementation and evaluation baseline for mapless autonomous navigation using PPO [1] and domain randomization [2], [8]. By coupling sector min-pooled LiDAR readings with dead-reckoning goal tracking, the policy achieves high navigation reliability under nominal simulation conditions (94%--98% success). Methodological rigor demonstrates that observed preliminary differences between nominal and randomized policies are not statistically significant at N=50, establishing a clear requirement for expanded benchmarking and perturbed-environment cross-evaluations. The completed architecture provides a solid trajectory toward completing the planned ablation matrix and physical robot deployment.

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